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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: R26 – DS - 002

Project Title: Interactive Sinhala Text Reader for the Visually Impaired Using OCR, NLP, AI.

1. Introduction

1.1. Background:

Visually Impaired Sinhala readers have low/no access to information through documents such as newspapers, textbooks, articles, etc. While there are devices that exist to help with these, there are not available in the Sinhala script and overall, the support is very low.

1.2. Research Problem:

Current technologies fail to recognize stylized headline typography, correct OCR-focused errors, and personalize voice input and generation specific to Sinhala.

1.3. Objectives:

It Number	Objective	Objective number
IT22072238	Develop a font-invariant Sinhala Character Recognition using Medial Axis Transform	1
IT22259134	Design and develop a context-aware two stage post-OCR error correction pipeline for Sinhala text.	2
IT22120670	Design and develop a RAG-based pipeline that dynamically simplifies or elaborate content in Sinhala, that adapts to user comprehension needs.	3
IT22004468	Design and develop Personalize and Voice Interaction Module that enables natural Sinhala speech interaction, fine-tuned intent classification and behavioral profiling to dynamically personalize responses for visually impaired users.	4

2. Data Exploration

2.1. Data Collection

Component 1: Collections of .ttf files were saved from various websites and manually filtered out, resulting in 295 distinct fonts. With use of python libraries and knowledge of wijesekara keyboard mappings, legacy fonts mapping and character classes were able to create ~300 classes of with 295 images of each (not accounting for any pre-training augmentations).
Furthermore for testing and developing the pipeline, I made use of the publicly available SinOCR dataset, which may be taken into consideration for future training

Component 2: 138 Lankadeepa,Mawbima,dinamina newspaper pages were photographed and passed through Tesseract OCR. OCR output was aligned with the original digital text to produce 1,211 real noisy-clean pairs, with 1883 errors manually classified into 5 categories for the error taxonomy. For training, 31,609 clean Sinhala sentences were scraped from Lankadeepa and Mawbima digital portals. A noise engine injected five OCR error types (calibrated to real frequencies) to produce 90,000 synthetic pairs. An additional 30,000 pairs came from RoundTripOCR - clean text rendered as images and OCR's back. All text was Unicode NFC normalized and invisible characters stripped before use.

Component 3: 5,000 clean Sinhala articles were scraped from Lankadeepa, Mawbima, Dinamina and Ada Derana digital news portals and collected from school textbooks using BeautifulSoup. These articles span multiple domains including news reporting, educational content, and formal government language from gazettes. The scraped text was Unicode NFC normalized and stored with raw and clean text fields to serve as the retrieval knowledge base for the RAG pipeline.

Component 4: Three data sources were used for this component.

- Data Source 1 - CLINC150 English Intent Dataset: The CLINC150 benchmark dataset was sourced from HuggingFace Datasets. It contains 22,500 labeled English

utterances across 150 intent classes. Relevant intent categories were filtered to match the assistive reading domain, specifically: summarize, explain, simplify, elaborate, rephrase, and identify content. This dataset was selected because the NLLB-200 translation model converts all Sinhala voice commands to English before classification, allowing a high-quality English intent dataset to be used in place of an unavailable Sinhala NLU dataset. This directly addresses the low-resource Sinhala NLP challenge identified in the research proposal.

- Data Source 2 - Sinhala Voice Recordings (STT Evaluation): Six real human voice recordings were collected manually in Sinhala, covering varied command lengths, hesitation patterns, and speaking styles. An additional set of AI-generated Sinhala audio samples was produced using edge-tts for supplementary testing. These recordings were used to evaluate the Speech-to-Text pipeline by measuring Word Error Rate (WER) and Character Error Rate (CER) across five STT models: Whisper base, Whisper small, Whisper medium, Wav2Vec2-BERT, and the Google SpeechRecognition Library (selected).

- Data Source 3 - Synthetic User Interaction Logs (Behavioral Profiling): A Python script was used to programmatically generate 500 synthetic user interaction log records representing realistic usage patterns across four distinct user personas: Students, Elderly Users, Professionals, and Casual Users. Each record captures the user's command, detected intent, personalization flags (speed, detail_level, language_style), and session metadata. These logs are stored in MongoDB Atlas and serve as the training dataset for the Random Forest behavioral classification model. This synthetic generation approach was chosen due to the absence of existing Sinhala assistive technology usage datasets.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
.tff files	Contains various fonts files used in Sinhala.	Multiple Webpages on Sinhala fonts/letters	28mb (extracted to trainable images, it was ~500mb)	These images were skeletonized and used in the training process.
SinOCR and SinFUND	Sentence and word level images in different fonts with the labels	SinOCR	1.4GB	These were used to test tje pipeline
Photographed newspaper pages	Real OCR pairs from 38 newspaper pages aligned with ground truth	Tesseract OCR	11,211 pairs	noisy_text, clean_text, error_type
Scraped clean corpus	Clean Sinhala articles from digital news portals	BeautifulSoup scraper	31,609 sentences	sentence_id, text, source

Synthetic noise pairs	Clean corpus with programmatically injected OCR noise	noise_engine	90,000 pairs	noisy_text, clean_text, error_type
RoundTripOCR pairs	Clean text rendered as image -> OCR's back -> aligned	PIL + Tesseract	30,000 pairs	noisy_text, clean_text
Sinhala newspapers, school textbooks and legal documents (gazettes)	Clean text Sinhala paragraphs	Newspapers, School Textbooks, Government gazettes	600 paragraphs	Raw_text, Clean_Text, Label
Scraped clean Sinhala Newspaper articles and Schools textbooks contents	Contains the detailed clean Sinhala articles that used as Knowledge base to RAG	Newspapers, School Textbooks	5000 articles	Raw_text, Clean_text
Sinhala Voice Recordings	Real and AI-generated Sinhala audio samples for STT pipeline evaluation	Manual collection	6 real + AI-generated samples	audio_file, sinhala_transcript, wer, cer, model_tested
CLINC150 Dataset	Labeled English intent utterances filtered for assistive reading domain intents	HuggingFace Datasets	22,500 utterances (filtered subset used	utterance, intent_label, domain, split (train/val/test)
User Interaction Logs	Programmatically generated user interaction records representing 4 user personas for behavioral profiling model training	Python script + Mongo DB	500 records	user_id, intent, personalization_flags {speed, detail_level, language_style}, session_id, timestamp, persona_label

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

	1	2	3	4
Data source 1	Used to MAT skeletonized and train the model.	Real Sinhala OCR errors used for testing and error analysis	Serves as the primary knowledge base for the RAG retrieval pipeline. The 5,000 articles provide broad domain coverage across educational, journalistic, and formal register text, enabling retrieval of contextually appropriate chunks for both simplification and elaboration tasks.	Used to evaluate and select the STT pipeline component. WER and CER metrics across five models determined Google SpeechRecognition (si-LK) as the best-performing model with 0.00% WER.
Data source 2	Use to skeletonize and test the pipeline.	Clean Sinhala text source for generating large synthetic datasets		Directly supports intent classification model fine-tuning. English utterances align with the translated output of NLLB-200, enabling XLM-RoBERTa to be trained on high-quality labeled data without requiring a Sinhala NLU dataset.

Data source 3		70% of training data with realistic OCR noise injection		Serves as the training and evaluation dataset for the Random Forest behavioral profiling model. Feature vectors extracted from interaction logs map to user personas, enabling the system to dynamically personalize responses
Data source 4		Simulates real OCR artefacts to improve model robustness.		

3. Methodology

3.1. Data Preprocessing:

Objective 1:

The true font files were extracted to each possible letter, combination (letter + diatrics such as Halkirima, papilla and ispilla) as well as diacritics alone. These were sent to the MAT skeletonization pipeline which converted it to a 512 x 512 image that was a single pixel wide skeleton of the original. Also, use the SinOCR dataset to create and test the pipeline with same MAT skeletonization added as well as a plan to incorporate Dilation as well.

	Transformation technique		
	Extraction	Skeletonization	Finalizing
Data source 1	Accessed internal file through libraries and emulated wijesekara mapping to create printed dataset of 512 x 512 Images.	Used Medial Axis Transform to convert them to 1 pixel skeleton.	Organized into folders and classes for training with appropriate labels. Added augmentation at training stage.
	Segmented	Skeletonization	Dilated

Data Source 2	Pipeline segmented into words and then letters.	Skeletonized before sending the model to classify.	The 1px skeleton was dilatated for uniformity as well as providing more presence of pixels to the project.

Objective 2:

	Transformation technique			
	Data Cleaning	Data Normalization	Data Integration	Feature Engineering
Photographed newspaper pages	Removed soft hyphen	Unicode NFC normalization	Aligned with ground truth using difflib; merged into test set	Per-token CORRECT/ERROR labels from word level
Scraped clean corpus	Removed duplicates and malformed sentences, stripped HTML tags	Unicode NFC normalization	Merged into unified training set with synthetic and RoundTripOCR pairs	-
Synthetic pairs	Removed 18,086 pairs identical after Unicode cleaning	Unicode NFC normalization on both noisy and clean sides	Combined with RoundTripOCR for training split	[ERROR]/[/ERROR] markers inserted around flagged tokens. `correct:` prefix added for mT5
RoundTripOCR pairs	Removed pairs where OCR output was identical to input	Unicode NFC normalization	Kept separate from test set, combined with synthetic for training	Per-token CORRECT/ERROR labels from difflib alignment

Objective 3:

	Transformation technique			
	Data Cleaning	Data Normalization	Data Integration	Feature Engineering

Scraped clean Sinhala articles	Removed duplicate articles and malformed sentences; stripped HTML tags and whitespace	Unicode NFC normalization	Articles chunked at sentence/paragraph level and indexed into vector store for retrieval	Complexity label assigned to each retrieved chunk via SVM classifier (Simple/Medium/Complex) for re-ranking

Objective 4:

The three datasets underwent separate preprocessing pipelines tailored to their specific roles in the component. CLINC150 was filtered by intent domain, tokenized using the XLM-RoBERTa tokenizer, and label-encoded for classification. The Sinhala voice recordings were converted to 16kHz mono WAV format, and transcripts were Unicode NFC normalized

before WER/CER computation. The synthetic interaction logs were feature-engineered into numeric count vectors per user session before Random Forest training.

	Data Filtering	Data Cleaning	Data Encoding	Feature Engineering	Data Type Conversion
CLINC150 Dataset	Filtered to 6 relevant intent classes from 150	Removed out-of-scope utterances; Unicode NFC normalization	Label encoding of intent classes; XLM-RoBERTa tokenization with padding and truncation	Train/val/test split (80/10/10)	Text strings to token ID tensors
Sinhala Voice Recordings	—	Audio resampled to 16kHz mono WAV; transcript Unicode NFC normalization	—	WER and CER computed per sample and per model	Audio files to text transcripts
Synthetic Interaction Logs	—	Removed duplicate session records; validated JSON schema	Label encoding of persona classes (0–3)	Count-based feature extraction: summarize_count, explain_count, simplify_count, slow_speed_count, simple_style_count, session_count	JSON log records to numeric feature matrix (DataFrame)

3.2. Scalability

Component 1: The extracted font dataset is scalable with every new font that gets released, and while we trained on ~300 classes consisting of the most common letters, combination and then a few mandatory classes, theses can be increased to encompass much more rare characters as well. The SinOCR dataset has 90,000 images and testing as conducted on random samples of these.

Component 2: The 97,564 training pairs are sufficient for mT5-small. The noise engine can generate more pairs from any clean Sinhala corpus. The real OCR test set (1781 genuine error pairs) could be expanded with more newspaper sources to improve generalizability. The pipeline scales to any document length through sentence level chunking.

Component 3: The 5,000 articles knowledge base is sufficient for the current RAG pipeline scope and is directly scalable by ingesting additional scraped articles from any Sinhala digital source without architectural changes. The vector store index can accommodate significantly larger corpora, and the SVM-based complexity re-ranker generalizes any new chunk provided it contains Sinhala Unicode text. Expanding additional genres such as legal documents or academic texts would further improve elaboration quality for formal register queries.

Component 4: The CLINC150 dataset is scalable by design - additional intent classes can be added, and the XLM-RoBERTa model can be fine-tuned incrementally without retraining from scratch. The synthetic interaction log dataset is fully scalable as MongoDB Atlas automatically accommodates growing log volumes; additional user personas can be defined, and the Random Forest model retrained with minimal overhead.

3.3. Feature extraction

Component 1: After extracting and generating the characters in each file into 512 x 512 images based on combinations found through legacy mappings. After binarization and filtering, the medial axis transform was applied to achieve the 1-pixel skeleton to train the model. Generation itself produces a class list and organizes all images and respective labels.

Component 2: SinBERT extracts a per-token ERROR probability using full sentence context via bidirectional attention. A confidence threshold converts this to a binary CORRECT/ERROR label. mT5 uses the [ERROR] marked sentence as input. the markers serve as position aware features directing the model to correct only the flagged positions while preserving all other tokens.

Component 3: Retrieved text chunks from the knowledge base are passed through the SVM complexity classifier pipeline to extract the seven linguistic features : ASL(Average Sentence Length), AWL(Average Word Length), TTR(Type Token Ration), FCD(Formal Connective Density), FVER(Formal Verb Ending Ratio), PR(Punctuation Ratio) and LWR(Long Word Ratio) as defined in the feature engineering module. These features are used to assign a complexity label (Simple, Medium, or Complex) to each retrieved chunk. The SVM classifier, trained to 92.17% test accuracy, then re-ranks retrieved chunks so that the RAG pipeline selects content at the complexity level most appropriate to the user's comprehension needs before passing it to the LLM for simplification or elaboration.

Component 4: Three distinct feature extraction processes were applied across the three data sources. For the CLINC150 dataset, utterances were tokenized using the XLM-RoBERTa multilingual tokenizer, producing input_ids, attention_mask, and token_type_ids tensors used as input features to the fine-tuned classification head. For the Sinhala voice recordings, Word Error Rate (WER) and Character Error Rate (CER) were computed per sample as evaluation metrics, with audio length and hesitation pattern type noted as descriptive features. For the synthetic interaction logs, six behavioral count features were extracted per user session: summarize_count, explain_count, simplify_count, slow_speed_count, simple_style_count, and session_count. These six features form the input feature vector to the Random Forest classifier for user persona prediction.

4. Modelling and Results

Objective 1:

The model and pipeline achieved a CER of 35.3% again a CER of 51.9% from Tesseract. This is under the fact that the dataset contained unrealistically complex fonts, hence why both results had a higher-than-average CER rate. Furthermore, the pipeline created is a simple segment-based OCR system while tesseract uses a much more complex CNN + LSTM model and since both were tested on a completely different set of data (sinOCR), the results can be satisfactory for this stage.

I intend to make more use of the SinOCR dataset as well, by resizing and applying MAT

and dilating it furthermore for uniformity to train another model that will incorporate not only CNN but also LSTM.

Objective 2:

System	CER	WER
B1 - Raw OCR	0.0367	0.1048
B2 - Rule-based	0.0367	0.1048
B3 - mT5 only	0.0104	0.0249
Proposed	0.319	0.0862

B3 achieves lower CER but modifies every sentence without error gating. For a voice assistive system where, wrong words are spoken aloud, the proposed system's SinBERT-based gating is safer and provides per-token confidence metadata to Component 3.

Objective 3: The RAG pipeline successfully retrieves contextually relevant Sinhala text chunks from the 5,000 article knowledge base and re-ranks them by linguistic complexity using the SVM classifier. The complexity re-ranking stage leverages the trained model's 92.17% classification accuracy to ensure retrieved chunks align with the target user's comprehension level prior to LLM-driven simplification or elaboration. End-to-end pipeline evaluation metrics including BLEU, ROUGE, and human readability scores are currently pending and will be reported upon completion of the full pipeline integration.

Objective 4: The component pipeline consists of three trained models working in sequence. In the STT stage, Google SpeechRecognition Library (si-LK) achieved 0.00% WER on 6 out of 7 evaluation samples, outperforming all four alternative models tested (Whisper base/small/medium and Wav2Vec2-BERT). For intent classification, the NLLB-200 translation + Ollama Llama 3.2:1b baseline achieved 85.7% accuracy on 7 evaluation samples. The XLM-RoBERTa model fine-tuned on CLINC150 is expected to exceed this baseline accuracy with lower latency. For behavioral profiles, given a clear separation between the four defined individual clusters, the random forest classifier trained on 500 synthetic interaction log records is expected to achieve an individual classification accuracy of over 90% on the retained test data.

4.1. Key Insights:

Component 1:

- Character analysis shows technically feasible combinations exceeding 700+ character combinations. However, with the help of a corpus, it was found that ~300 classes represent 99.7% of all characters, highlighting the extreme rarity of certain sets of characters.
- Feature reduction helps reduce data dimensionality by 70%, which is a goal to reduce the invariance and invariance of features in font text.
- Unlike English where simply letters exist as combination sides by side, in Sinhala, the letter data is split into standard single letters, letters with in-built diatrics (Hal-kirima Papilla, rakaranshaya), left-sided diatrics (kombuwa/dheka) as well as right side diatrics(Alapilla,adhaya...). This gave complexity but also helped define better logic in character recognition.

Component 2:

- Manual annotation of 1883 errors showed vowel modifier omission is the dominant OCR error type at 46.9%, confirming that Sinhala OCR errors are primarily sub-word level. this is why CER was chosen as the primary metric over WER
- SinBERT achieved F1=0.9801 at threshold=0.65, showing that a Sinhala-pretrained contextual model can reliably detect OCR errors using sentence context. Higher thresholds (0.85–0.95) further reduced over-flagging in real-world conditions.
- The proposed pipeline achieved CER=0.0274 and WER=0.0752 on 1,211 real test pairs. Although B3 achieved lower CER=0.0065, it lacks error gating and confidence metadata. essential for a voice assistive system where over-correction causes wrong words to be spoken aloud.

Component 3:

- The 5,000 article knowledge base provides sufficient domain diversity across journalistic, educational, and formal Sinhala registers to support both simplification and elaboration tasks across a wide range of user queries.
- Integrating the SVM complexity classifier as a re-ranking stage within the RAG pipeline ensures that retrieved chunks are matched to the user's comprehension level before generation, directly addressing the adaptive content delivery objective.

- ASL (Average Sentence Length) was identified as the dominant complexity signal, meaning the re-ranker is particularly sensitive to sentence length differences between Simple, Medium, and Complex Sinhala passages a linguistically meaningful property for readability adaptation.

Component 4:

- The key insight from the STT evaluation is that lightweight, language-specific models outperform heavy general-purpose models for low-resource languages: Google Speech Recognition with si-LK language code achieved near-perfect transcription on Sinhala while all Whisper variants produced hallucinations and repeated-word loops.
- The key insight from the intent detection comparison is that the NLLB translation bridge is essential — the direct Sinhala approach (Qwen2.5:1.5b) achieved 0.0% accuracy without translation, confirming that the two-step NLLB + classifier pipeline is the correct architectural choice for low-resource Sinhala NLU.
- The key insight from the behavioral profiling design is that six simple behavioral count features are sufficient to distinguish user personas without requiring complex deep learning, making the Random Forest approach computationally appropriate for a CPU-only assistive device.

4.2. Challenges Faced During Data Analysis:

Objective 1:

Understanding how true font files work, especially in Sinhala as the mapping is completely different and much complex compared to English and Sinhala Unicode (these fonts use a legacy mapping system). The extraction was a bit complex to size as needed. The skeletonization was challenging, and various fields were adjusted to make sure all font images produce a consistent skeleton under the same parameters.

Objective 2:

Invisible Unicode (ZWJ/ZWNJ) causing loss=0.0 (NaN) --> clean_sinhala() strips all invisible characters; token-level identity check applied

SinBERT labels causing training collapse (59% flagged vs 26% GT) --> Switched to difflib ground truth alignment for all training labels

Objective 3:

The primary challenge was the absence of a preexisting labeled Sinhala readability corpus suitable for RAG re-ranking. This was resolved by constructing a 600-paragraph annotated dataset from newspapers, school textbooks, and government gazettes, and training a dedicated SVM complexity classifier to assign complexity labels to retrieved chunks at inference time. A secondary challenge was ensuring consistent Unicode representation across scraped sources, resolved through NFC normalization applied uniformly across all text prior to chunking and indexing.

Objective 4:

The primary challenge was the absence of labeled Sinhala NLU datasets for intent classification. This was resolved by using NLLB-200 as a language bridge to translate Sinhala to English, enabling use of the established CLINC150 English intent benchmark. A second challenge was hardware constraints made running large local LLMs slow Ollama Llama 3.2:1b produced a 41 second response on first call and 4-15 seconds on subsequent calls - reinforcing the decision to replace the Ollama approach with a lightweight fine-tuned classifier.

5. References

[1] UNPRPD, "Situational Analysis of the Rights of Persons with Disabilities in Sri Lanka," Jan. 2025.

[2] T. Madhusha et al., "Mobile Base Sinhala Book Reader for Visually Impaired Students, 2023

[3] NLPC-UOM, "SinBERT," *arXiv*, 2022.

[4] L. Xue et al., "mT5," *Proc. NAACL*, 2021.

[5] R. Smith, "Tesseract OCR Engine," *Proc. ICDAR*, 2007.

[6] S. Weerasinghe et al., "Post-processing of Sinhala OCR," *Proc. IEEE ICIAfS*, 2016.

[7] S. Premasiri et al., "Morphological Dictionary Post-OCR Correction," *Proc. ICAC*, 2018.

[8] A. Kashid et al., "RoundTripOCR," *Proc. LREC*, 2022.

[9] A. Raffel et al., "T5," *JMLR*, vol. 21, 2020.

[10] M. Suresh, R. Vijaya Kumar, and A. G. Ramakrishnan, "A Medial Axis Based Thinning Strategy for Character Images," 2016.

[11] S. Cheng, Y. Ren, and Y. Yang, "SSR-GNNs: Stroke-Based Sketch Representation with Graph Neural Networks," (CVPRW), 2022

[12] L. Lam, S.-W. Lee, and C. Y. Suen, "Thinning Methodologies — A Comprehensive Survey," Sep. 1992.
[13] Researchers, University of Moratuwa, "Benchmarking OCR Models for Sinhala and Tamil Document Digitization," in Proc. Engineering Research Unit (ERU), 2024.
[14] N. Wijesekara and R. Ranathunga, “Applicability of end-to-end deep neural architectures for Sinhala speech recognition,” in Proc. International Conference on Advances in ICT for Emerging Regions (ICTer), Colombo, Sri Lanka, 2018.
[15] A. P. Karunanayake, “Sinhala speech recognition for interactive voice response systems,” M.Sc. thesis, Univ. of Moratuwa, Sri Lanka, 2017.
[16] A. Gupta, R. Kumar, and S. Sharma, “Intent recognition framework for low resource spoken dialogue systems,” arXiv preprint arXiv:2104.08755, 2021.